

TABLE II
Colour Codes of the Wires
for 55SIM-25DAYG
Stepper Motor

Colour	Identification
White	Common
Red	L1
Blue	L2
Brown	L3
Yellow	L4

55SIM-25DAYG unipolar stepper motor rated at 12V DC, with step angle of 7.5° per pulse. It is shown in Fig. 5. This stepper motor has five wires for connection to the circuit. Out of the five wires, four are coming out from the coils of the stepper motor and the fifth is the centre-tapped common terminal.

Correct identification and proper connection of these wires are important for successful assembly of the stepper motor. The unipolar stepper motor has two centre-tapped coils. Their centre-tapped terminals are joined to make one common terminal. If you have a multimeter, adjust it in the 200-ohm range and check resistances between wires. The resistance between the common terminal and the rest of the coil terminals should be about 36 ohms, while the resistance between any two coil terminals should be about 72 ohms. Connect the coils to the output terminals of the driver circuit in sequence.

To make the project more user-friendly, colour codes of wires for the 55SIM-25DAYG stepper motor are shown in Table II. Note that for the same ratings, different manufacturers may have their own coil specifications.

Matlab program

This program consists of 14 M-files and four figure files. However, only three main programs (parport.m, data_port_controller.m and data.m) have been explained and listed here. Matlab help files are very comprehensive. To get general help, type 'doc' command against '>>' prompt in 'command' window.

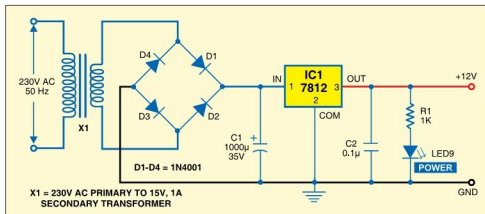


Fig. 4: Circuit for 12V power supply



Fig. 5: The stepper motor

or LPT3 only. LPT1 corresponds to dataport address 0x378.

Typing 'dio=digitalio('parallel', 'LPT1')' on Matlab command line gives an output as shown in Fig. 6. 'addline' command adds hardware to the digital I/O object. The syntax is:

```
line = addline(obj, hwline, port,
'direction')
```

The 'putvalue(data, 1)' command makes LED1 glow.

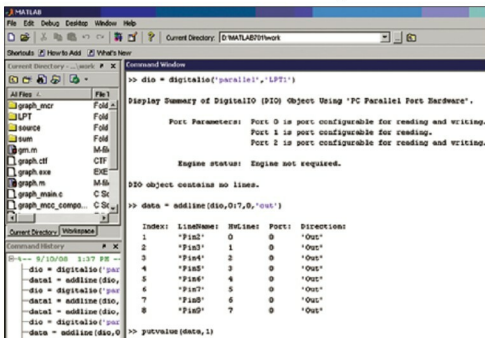


Fig. 6: Screenshot of Matlab command window

window.

Matlab has a toolbox called DAQ (data acquisition). The 'dio = digitalio('adaptor', ID)' creates the digital I/O object (dio) for a specific adaptor and for the hardware device with device identifier ID defined with an integer or a string. The ID can be LPT1, LPT2

The above codes are defined in 'parport.m' program as given below:

```
dio = digitalio('parallel', 'LPT1');
data = addline(dio, 0, 7, 0, 'out');
putvalue(data, value); %where value
b/w 0 and 255
```

Matlab comes with a 'Guide' tool. Guide (short for 'graphical user